

14 – Technical Foundations

The planning basis engineering inherits: architecture posture and its rationale, the technology stack, the till platform, a provider-neutral hosting and capacity model with the hosting decision framed as a matrix, the standards roadmap by phase, the integration seams, the assurance sequence that retires the largest technical risks before build, and the register of technical decisions resolved and open.

Field	Value
Document	14 – Technical Foundations (planning basis; Stage 1 of the design phase; number reserved in the register since August 2026)
Version	v0.2 – draft for design review (adds section 1.3, stateless scale-out)
Date	7 September 2026
Status	Draft. Written provider-neutral by decision: hosting is chosen from the matrix in Section 5 once quotes are in; the till terminal model is chosen after the Section 8 spikes. Nothing downstream should cite a provider or a terminal brand.
Inputs	06 NFR Specification; 09 Decisions Log (ADR-10, 19–25); 16 Questionnaire Answers (H, Q, R, S); 18 Core Data Model; 19 Cross-Cutting Services; 20 Stage 2 plan v0.2; 30 Frontend Foundations; 31 Update and Distribution; the August 2026 technical discussion (stack fit, hosting cost basis, Android direction) and the September 2026 review decisions.
Audience	Engineering leads, DevOps, hardware procurement, the federation technical reviewer. Sections 1, 5 and 9 for programme readers.

What this document is not. It is not the solution architecture (doc 17 owns module boundaries and the event backbone) and not the deployment design (doc 35 makes the chosen hosting concrete). It is the floor both of those stand on: the choices that are stable across providers and vendors, the numbers that size everything, and the decisions that must close before Stage 2 week 3.

1. Architecture posture

Six properties of the problem decide the posture. Each maps to one architectural commitment, and the commitments are deliberately few.

Property of the problem	Commitment	Where it is designed
876 legal entities trading with each other; one federation view	One shared database, rows owned by entity, row-level security; no per-tenant schemas	Doc 18 section 3.7
Ownership and risk transfer at GRN; documents are legal records	Immutable documents, gapless series, compensating corrections, append-only ledgers	Doc 18 parts C and D
Tills must trade 1–2 days offline on poor links	Offline-first till with a full local database; one writer per record type; sequence-ordered idempotent sync; no merge logic	Doc 19 section 6, doc 32
Single DevOps engineer operating model	Modular monolith, one deployable, managed or simple services, no orchestration platform in phase 1	This document, doc 20 section 4
10,000 shops, 3 million receipts a day at the horizon	Partitioned ledgers, outbox-fed projections on a replica, ingest designed for reconnection bursts	Section 6, doc 19 6.2
Trilingual, low computer familiarity	Screens as paper; ICU collation and normalisation; raster receipts; three-column names with English anchor	Doc 19 section 5, doc 30

1.1 Modular monolith, enforced

The application is one Spring Boot deployable organised as modules with build-time boundary enforcement (Spring Modulith): a module's tables, services and events are its own; other modules use its published commands and queries or react to its events through the transactional outbox. The enforcement is what keeps a monolith from rotting into a tangle; it also keeps extraction into a separate service possible later, which is the answer to “what if one module needs to scale alone” (doc 20 Appendix A).

1.2 Portability as the hedge

Because hosting is undecided and may change at the pilot gate, every infrastructure dependency is chosen for portability: PostgreSQL, an AMQP or NATS broker, S3-compatible object storage, OCI containers, OpenID Connect. Moving between a cloud provider and local hosting is then a deployment exercise (doc 35), not a redesign.

1.3 Stateless scale-out: any request to any instance

Application instances hold no request-affinity state, so an instance can be added, removed or replaced at any moment and the load balancer needs no sticky sessions. A request from a user or a till that last spoke to instance A is served identically by instance B. The rules that make this true:

Concern	Rule	Mechanism
HTTP session	None. No server-side session store; Spring Security runs stateless.	Identity, scope, roles version and MFA freshness travel in the access token (doc 19 section 1); refresh tokens live in the identity provider, not the application.
Permission and reference caches	Per-instance in-memory caches keyed by catalogue version; never authoritative.	Invalidation by a version-bump event fanned out through the broker; a new instance warms from the database on start. No shared cache service in phase 1.
Till sync	Every upload batch is self-contained and idempotent; the acknowledgement is derived from database state, not instance memory.	The device's last applied source_seq lives in a row locked for update during ingest; the till keeps one batch in flight, so two instances can never interleave one device's sequence.
Command retries after failover	A retried command must not execute twice.	Idempotency key header on every mutating web command, recorded with the result in the database; the same key returns the same result from any instance.
Singleton work	Outbox relay, end-of-day aggregation, gap checks, expiry tasks run once, not once per instance.	Clustered scheduler locks (database advisory lock or ShedLock class); the relay may be partitioned by source for throughput, each partition singly owned.
File and image uploads	Never touch instance disk.	Streamed to object storage; large uploads use pre-signed URLs from the device straight to storage.

Concern	Rule	Mechanism
Push to web clients	No instance-bound connections in phase 1.	Polling with stale-while-revalidate; if server-sent events are added later, each instance subscribes to the broker so any instance can deliver.
Database connections	Adding instances must not exhaust PostgreSQL connections.	A connection pooler (PgBouncer class) in front of the primary; per-instance pool size × instance count stays under the pooler's ceiling.
Load balancer	Layer-7, least-connections or round-robin, health-checked, connection-draining on removal, TLS terminated at the edge.	Readiness endpoint reports true only after caches are warm and the database is reachable; removal drains in-flight requests before the instance stops.
Proof	Instance-affinity bugs are caught before production.	Development, CI and the pilot always run at least two instances; a test suite drives one client across both.

The scaling ceiling is therefore the database, not the application: writes scale vertically and by partition pruning, reads by the reporting replica. That is the intended shape for a single-DevOps model; the application tier is the cheap, elastic part.

2. Technology stack

Layer	Choice	Rationale and constraints
Backend language and framework	Java 21, Spring Boot 3, Spring Modulith	Team experience (Java/Spring); Modulith enforces the module boundaries of doc 20 section 2.1; long-term support releases only.
Database	PostgreSQL 16+	ICU collations for Sinhala and Tamil, row-level security, range partitioning, exclusion constraints for effective-dated uniqueness, logical replication for the reporting replica. Schema migrations with Flyway, forward-only, expand-then-contract.
Broker	RabbitMQ (quorum queues) by default; NATS JetStream as the lighter alternative	Fan-out and buffering only; the outbox retains (doc 19 DR-2). Kafka deferred.
Identity provider	Keycloak-class OIDC provider, self-hosted, single realm	Doc 19 DR-1; TOTP, client-certificate device auth, admin API wrapped behind the scope rule.
Object storage	S3-compatible bucket service	Attachments, images, release packages, audit hash anchors (write-once bucket).
Web back office	React 18+, TypeScript, Vite; headless component primitives styled by the design system	Doc 30 section 3.
Till application	Kotlin, Jetpack Compose, Room/SQLite with encryption, WorkManager for sync	Doc 30 DR-1: native for device-owner control, peripherals, fsync-controlled local database, raster printing.
Driver application	Same Android stack, minimal	Doc 30.
API	REST with OpenAPI 3.1; generated typed clients	Doc 29 / doc 33.
Rendering	Server: PDF pipeline with embedded Noto fonts and HarfBuzz shaping; till: platform text stack to raster	Doc 19 5.3.
Search	PostgreSQL trigram plus ICU collations (phase 1); OpenSearch analysers (phase 2)	Doc 19 5.4.
Edge	A CDN/WAF/TLS edge service in front of the load balancer (provider-neutral)	DDoS absorption and TLS termination; the tills tolerate edge outages by design.
Runtime and deployment	Containers on virtual machines behind a load balancer; two or more application instances; Ansible or equivalent for configuration; no Kubernetes in phase 1	Single-DevOps model. A managed container platform is acceptable if the chosen provider offers one that needs no cluster operations; otherwise plain VMs (Section 5).
Observability	Metrics, logs and traces to a self-hosted or managed stack (Prometheus/Grafana/Loki class); crash reporting for the till	Doc 36.
Secrets and keys	A key-management service or a self-hosted vault; release signing key under dual control	Doc 31 section 8, doc 34.
Device management	Device-owner management agent (open-source MDM class) validated in the spikes; vendor console for firmware	Doc 31 DR-1.

3. Till platform

Direction confirmed: ARM-based Android terminals, all new, centrally specified (H-08, T-07). In practice this is the class of integrated Android POS terminals with built-in printer and battery. The consequences for design are already absorbed: the internal battery turns the power-cut requirement into a device property (Q-07); the integrated 80 mm printer takes raster output, which solves trilingual receipts; the peripheral abstraction is the platform's own interface over two or three vendor SDKs, with UnifiedPOS used only as vocabulary; updates are self-managed through a device-owner agent (doc 31).

3.1 Minimum terminal specification

Attribute	Minimum	Why
Platform	Android 11 or later, 64-bit ARM, device-owner provisioning supported (QR or zero-touch), lock-task mode	Doc 31; long OS support window
Memory and storage	4 GB RAM, 64 GB storage	Two-day local database plus catalogue thumbnails plus headroom for updates
Screen	8 inch or larger, capacitive, 1280 × 800 or better	Doc 30 zoning; 56 dp targets
Printer	Integrated 80 mm thermal, raster (bitmap) printing at 203 dpi, auto-cutter	Trilingual receipts; Z-reports offline
Scanner	2D imager (built-in or USB/serial), reads EAN/UPC, GS1-128, GS1 DataMatrix, QR	GS1 2027; batch resolution
Battery	Internal, 4 hours of trading on battery; mains adapter	Q-07
Ports	USB-A/serial for scale, cash drawer (RJ11 or USB), optional customer display, Ethernet or Wi-Fi plus 4G option	H-07; Q-06 failover
Security	Hardware-backed keystore; verified boot	PIN hashes, device credential, encrypted local database
Vendor	Local distribution, warranty and repair presence in Sri Lanka; 3-year parts availability; a published Android SDK for printer, scanner and drawer	15,000 units make logistics a design constraint

3.2 Peripherals outside the terminal

Device	Specification	Note
Weighing scale	Serial or USB, protocol documented, price-computing not required; label-printing weigh station optional for larger shops	The single riskiest integration (Section 8, spike 3); the scale, not the printer, decides the vendor
Cash drawer	Printer-driven kick or USB	Drawer-open-without-sale is an audited event (N-09)
Label printer	Where present: direct-thermal, 2D-capable; prints shelf labels, batch stickers, weigh labels	Optional per shop (B-11); MRP-wise stickers (multi-MRP option C)
Customer display	Terminal second screen or a small USB display	Doc 30 DR-4
Card terminal	Standalone bank terminal; no integration	ADR-19: reference recorded only
Router	Shop-provided per Q-06; minimum 2 Mbps down / 0.5 up; 4G failover recommended	Go-live gate per site

3.3 Platform choice: Google services or not

	GMS-certified terminal	AOSP terminal
Update path	Device-owner agent or Managed Google Play private app	Device-owner agent only
Developer verification (2026–27)	Exempt through the agent path; register anyway as a hedge	Out of scope
Background load and memory	Play services present	Leaner
Vendor ecosystem	Broader	Common among POS-specific makers
Decision	Not decided by software: the update path works on both (doc 31). Choose on hardware merits, SDK quality, local support and price.	

4. Standards roadmap by phase

Standard or regulation	Phase 1	Phase 2	Phase 3
OpenID Connect / OAuth2, TOTP	Yes		
OWASP ASVS Level 2; NIST SP 800-63B credential rules	Yes		
GS1 barcodes: EAN/UPC, GS1-128, in-store 20–29 prefixes; GS1 DataMatrix and Digital Link parsing (AI 01/10/17)	Parse and resolve GTIN; capture batch when present	Batch-resolved pricing where marked	
Unicode NFC; ICU collation and formatting for si-LK, ta-LK, en-LK	Yes	OpenSearch analysers	
WCAG 2.2 AA (web); Material accessibility guidance (till)	Yes		
OpenAPI 3.1	Yes		
IAS 2 weighted-average costing	Yes	Landed cost for imports	
VAT (standard rate; SVAT abolished 1 October 2025)	Yes		
Consumer Affairs Authority Act s.20(5) gazetted ceilings	Yes (hard block)		
Personal Data Protection Act No. 9 of 2022	Compliant by design		
Inland Revenue e-invoicing	Readiness (immutability, numbering, seam)		Live integration when mandated
Android Enterprise device-owner provisioning; developer verification	Yes		
Audit hash chain with write-once anchors; SIEM export	Reserved columns	Yes	
Columnar analytics store		When end-of-day exceeds its window	

5. Hosting: the decision matrix

Hosting is deliberately open (September 2026 decision). Three options are live. The design works on all three; the cost of each is not only the monthly bill but the operating model it forces. The matrix states what changes under each option so the decision is made with the consequences visible. Prices are indicative planning ranges from the August 2026 estimate and must be replaced by quotes.

Criterion	Cloud provider with managed services (DigitalOcean-class)	Hyperscaler (AWS-class)	Local hosting (colocation or Federation premises)
Indicative monthly run cost at pilot scale	Lowest of the cloud options; the August estimate placed it at roughly half to 60% of the hyperscaler figure	Highest; broadest managed catalogue	Capital purchase plus colocation, power, links and staff; lowest recurring cloud bill, highest fixed cost
Managed PostgreSQL with replica and point-in-time recovery	Yes, with cross-region replica options	Yes	No: the Federation runs Postgres, backups and failover itself
Managed broker, object storage, load balancer	Object storage and load balancer yes; broker self-run on a VM	All managed	All self-run
Operating model	Single DevOps engineer is credible	Single DevOps engineer is credible	Needs at least two operations staff on rota plus hardware support; ADR-20 must be revisited
RTO 4–8 h / RPO 15 min (ADR-21)	Achievable with managed backups and a standby region	Achievable	Achievable only with a second site (or a cloud standby) and rehearsed failover; otherwise RTO becomes days after a site loss
Data residency	Nearest regions are outside Sri Lanka; no legal constraint identified (Q-01) but the government IT policy check is still open	Same	In-country by construction; the strongest answer if the policy check turns out to require it
Latency for tills	Regional (South or South-East Asia) latency is acceptable; tills are offline-first	Same	Lowest, but irrelevant to trading (offline-first)
Portability	High: Postgres, S3-compatible, containers	High if the managed catalogue is used narrowly	High
Update distribution and fleet	Unaffected (doc 31)	Unaffected	Package store and update service on the local host; unaffected otherwise
Growth to 12,000 shops	Vertical then replica scaling; columnar store when needed	Same, with more options	Hardware refresh cycles must be budgeted
Main risk	Fewer managed pieces than a hyperscaler	Cost; over-use of proprietary services	Site loss, staffing, procurement lead time, power

Recommendation for the decision. Choose the cloud provider with managed Postgres unless the government IT policy check (Q-08, still open) requires in-country hosting. If local hosting is chosen for cost or policy, adopt a hybrid: primary in-country, backups and a warm standby in a cloud region, so the RTO/RPO commitment survives a site loss and the operating model stays close to single-DevOps. Decide at the pilot gate at the latest; nothing before pilot depends on the answer except doc 35.

6. Capacity model (planning basis)

All sizing derives from the stakeholder-confirmed envelope (ADR-22): 10,000 shops within three years, about 15,000 tills, about 3 million receipts a day; 12,000 shops at five years. The figures are order-of-magnitude planning values; spike 5 (Section 8) replaces them with measured ones.

Quantity	Planning value	Derivation
Receipts per day (horizon)	3,000,000	ADR-22
Lines per receipt	5 average	Grocery basket assumption
Receipts per second	About 100 average over a 9-hour trading day; 300 at evening and festival peaks	$3M \div 32,400 \text{ s}$; peak factor 3
Events per receipt reaching central	About 8 (document, lines, state row, movements, audit)	Doc 19 6.3

Quantity	Planning value	Derivation
Central ingest, steady state	About 1,000 events per second at peak	300 receipts × 8, batched per till
Reconnection burst after a wide 2-day outage	About 6 million receipts queued; catch-up target within 4 hours at about 3,500 events per second	H-01; ingest is rate-limited per device, the broker absorbs, projections lag with freshness stamps
Hot data growth	About 2.5 KB per receipt with lines; about 7.5 GB per day; about 2.7 TB per year before compression	Monthly partitions; cold partitions to cheaper storage
Audit rows	About 2 per receipt plus back-office events; about 6 million a day	Partitioned by received_at; indefinite retention (J-06) drives the tiering policy
Catalogue snapshot per shop	Under 20 MB including thumbnails for the shop's assortment	Doc 20 M2; delta pushes thereafter
Back-office concurrent users	About 2,000 at peak across 876 entities	Office staff estimate; low intensity

6.1 Sizing at pilot and at horizon

Component	Pilot (about 400 shops)	Horizon (10,000 shops)
Application instances	2 × 4 vCPU / 8 GB	4–6 × 8 vCPU / 16 GB; a second role for ingest-only instances during bursts
PostgreSQL primary	4 vCPU / 16 GB, NVMe	16 vCPU / 64 GB, NVMe; replica for reporting; partition pruning essential
Broker	1 × 2 vCPU / 4 GB	3-node quorum, 4 vCPU / 8 GB each
Object storage	Under 100 GB	Several TB (attachments, images, packages, cold partitions)
Identity provider	1 × 2 vCPU / 4 GB	2 × 4 vCPU / 8 GB
Reporting replica	Shared with primary	Dedicated replica; columnar store when end-of-day exceeds its window

7. Integration seams

Seam	Direction	Phase	Design
Accounting system	Out (journal export); later API	Integration when chosen	Posting maps as data per document type (J-02); OpenAPI consumer (R-07)
Inland Revenue e-invoicing	Out	P3	Reserved adapter; invoices already immutable and gaplessly numbered
SMS and email gateway	Out	P1	Provider-neutral adapter; templates in three languages (R-08)
Terminal vendor SDKs	In-device	P1	Behind the peripheral abstraction; two or three vendors at most
Device management agent	Central to device	P1	Doc 31
Object storage, key management, broker, edge	Infrastructure	P1	Portable interfaces (Section 1.2)
Department of Co-operative Development	Read-only role	P1	External time-boxed access class (M-10); no prescribed returns
Banks and payment networks	None	–	ADR-01, ADR-19

8. Assurance sequence: the spikes

Paper cannot settle these; code on real hardware must. Each spike has a pass criterion and feeds a decision. They run in Stage 3 alongside module design (doc 20 section 6).

#	Spike	Pass criterion	Feeds	When
1	Till local database and issuance under power loss	10,000 receipts issued with hard power cuts injected at random points: zero lost or duplicated numbers, database opens clean every time	Doc 18 5.5; doc 26	Weeks 2–3
2	Trilingual raster receipts	Golden receipts in Sinhala, Tamil, English and mixed fallback print correctly on each candidate terminal; Z-report offline	Doc 19 5.3; terminal choice	Week 3
3	Weighing scale and peripherals on candidate terminals	Scale reads reliably over the vendor's port; drawer, scanner, customer display and label printer function through the abstraction; simulator parity	Terminal choice (gating)	Weeks 3–4
4	2D scanning and GS1 parsing	DataMatrix and Digital Link resolve GTIN, batch and expiry; dual-marked packs beep once; 20–29 weigh labels decode	Doc 22, doc 26	Week 4
5	Sync load at the envelope	Simulated 15,000 tills: steady 1,000 events/s and a 6-million-receipt reconnection burst absorbed with projections lagging but no data loss; measured sizing replaces Section 6	Section 6; doc 35	Week 5
6	Device-owner management agent	Silent install, lock task, wake on push, resumable download, wipe, on both a GMS and an AOSP candidate	Doc 31 DR-1; terminal choice	Weeks 3–4
7	Cashier usability rounds	Real cashiers complete sale, refund, weigh and GRN flows on the prototype with the target tap counts and error rates in all three languages	Doc 30; T-04	Weeks 4 and 6

9. Technical decision register

9.1 Resolved (recorded here; ADR entries to follow in doc 09)

Decision	Resolution	Reference
Architecture style	Modular monolith with enforced boundaries, transactional outbox, one database	Section 1; ADR-20
Backend stack	Java 21, Spring Boot 3, Spring Modulith, PostgreSQL 16+, Flyway	Section 2
Web stack	React 18+, TypeScript, Vite	Doc 30
Till stack	Native Android (Kotlin, Compose), dedicated till track	Doc 30 DR-1

Decision	Resolution	Reference
Till platform	ARM Android integrated terminals, centrally specified, all new	Section 3; H-08, T-07
Identity	Self-hosted OIDC provider, single realm	Doc 19 DR-1
Broker role	Fan-out and buffering only; outbox retains; no log broker in phase 1	Doc 19 DR-2
Update distribution	Device-owner agent, signed packages, rings, health gates, version floor refusing sync not selling	Doc 31; ADR-24
Portability	Postgres, S3-compatible storage, AMQP/NATS, containers, OIDC; no proprietary managed service in the critical path	Section 1.2
Hosting neutrality	Design carries no provider; decision by matrix at the pilot gate at the latest	Section 5
Orchestration	No Kubernetes in phase 1; containers on VMs behind a load balancer	Section 2
Scale-out model	Stateless instances, no sticky sessions, idempotent commands, singleton jobs under cluster locks, connection pooler in front of PostgreSQL	Section 1.3

9.2 Open (decision requests)

#	Point	Proposed	Decider	Needed by
DR-1	Hosting option	Cloud with managed Postgres unless in-country is required; hybrid if local	Chamin, Federation	Pilot gate at the latest; earlier for doc 35
DR-2	Broker product	RabbitMQ quorum queues	Engineering lead	Stage 2 week 2
DR-3	Terminal model shortlist and final choice	Two or three candidates through spikes 2, 3, 6; scale integration is the gating test	Hardware procurement, engineering lead	Stage 2 week 4
DR-4	GMS or AOSP	On hardware merits; both supported	Hardware procurement	With DR-3
DR-5	Management agent product	Open-source device-owner MDM validated in spike 6	DevOps	With DR-3
DR-6	Archival and cold-storage tiering policy	Monthly partitions; partitions older than 24 months detached to object storage; indefinite retention pending J-06	Federation with tax advisor	Before pilot
DR-7	Columnar store timing	Trigger on end-of-day window; not phase 1	DevOps	Phase 2 planning
DR-8	Government IT policy check on data residency	Legal action item; outcome feeds DR-1	Federation legal	Before DR-1
DR-9	Observability stack product	Self-hosted Prometheus/Grafana/Loki class unless the provider offers an equivalent needing no operations	DevOps	Doc 36

10. Traceability

Section	Requirements and decisions
1 Posture	ADR-10, ADR-20, ADR-21, ADR-22; Q-02, Q-03; M section; doc 18 principles
2 Stack	NFR maintainability and security sets; P-04, P-05; R-07; ADR-20
3 Till platform	H-07, H-08, H-09; Q-06, Q-07; T-07; B-11; ADR-19; ADR-24
4 Standards	NFR security, localisation, compliance sets; J-03, J-04; G-03; K-08; M-09
5 Hosting	Q-01, Q-02, Q-03, Q-08; ADR-20, ADR-21; September 2026 hosting-neutrality decision
6 Capacity	ADR-22; H-01, H-06, H-12; Q-05; NFR performance and capacity
7 Seams	J-02, J-04; R-05 to R-09; M-10; ADR-01
8 Assurance	T-04, T-05, T-07; H-02; P-05; doc 20 Stage 3

End of document. Draft for design review; Sections 5 and 6 are replaced by quotes and measured figures as they arrive.